Assignment - 2

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Section A

Ans1 CPU registers are small, high speed storage locations inside the processors

importance :-

- store immediate data and instructions
- faster than cache and memory
- improve execution speed of programs

Ans2 Program counter is a register that holds the address of the next instruction

function :-

- points to next instruction to be executed
- automatically updates after each instruction
- control instruction sequence.

Ans3 Instruction format defines the structure of an instruction in binary form.

Key fields

- opcode
- operands
- addressing modes

Ans5 Addressing mode specify how to locate operands

Purpose

Provide flexibility in accessing data
Reduce instruction count
improve program efficiency.

Ans6 ISA is the interface between hardware and software defining in supported instructions.

includes

- instruction format
- Data types
- Registers and addressing modes.

Ans7 Types of instructions in a processor are:-

Arithmetic (ADD; SUB)
Logical (AND, OR)
Data Transfer (LOAD, STORE)
Control (JUMP, BRANCH)

Ans8 Explanation
- opcode is interpreted
- Requested operation is identified
- control signals are generated
- operands are prepared for execution.

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Ans 9 Basic instruction cycle are:-

Fetch instruction from memory
Decode instruction
execute operation
Store result

Ans 10 characteristics of RISC processors are:-

simple and fixed length instructions
few addressing modes
faster execution (pipeline friendly)
load/store architecture.

Ans 11 characteristics of CISC processors are

- complex and variable-length instructions
- many addressing modes
- fewer instructions per program
- slower but powerful instructions

Ans 12 RISC-V

- open source ISA
- simple fixed length instructions
- Easier to design and extend.

X86
→ proprietary (intel/AMD)
→ complex variable-length instructions.

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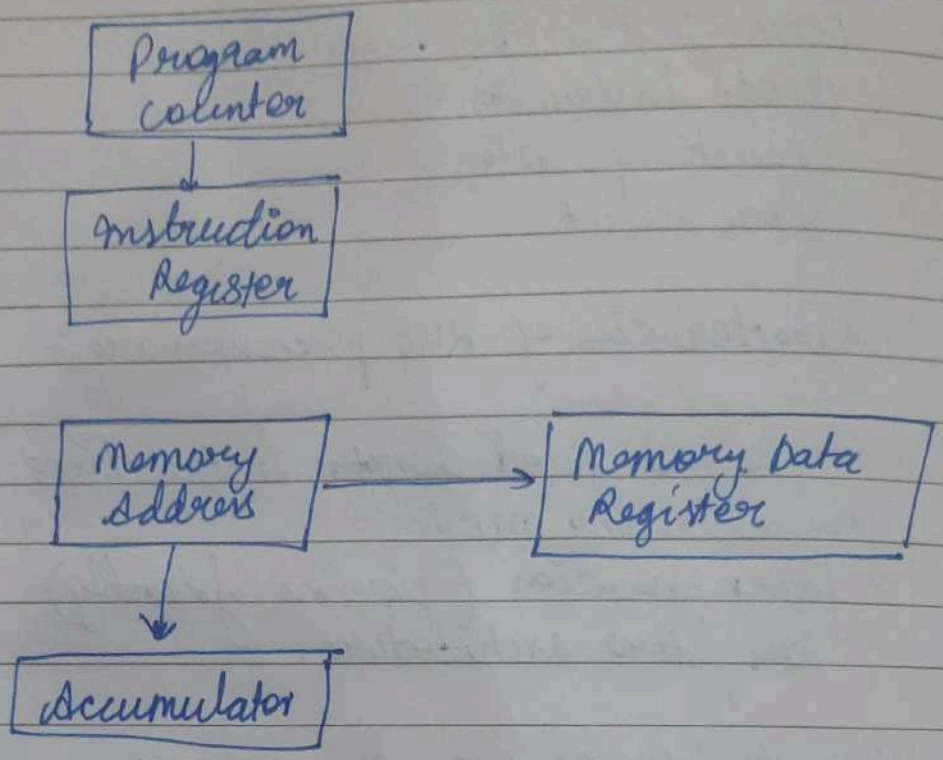
Section-B

Ans 1

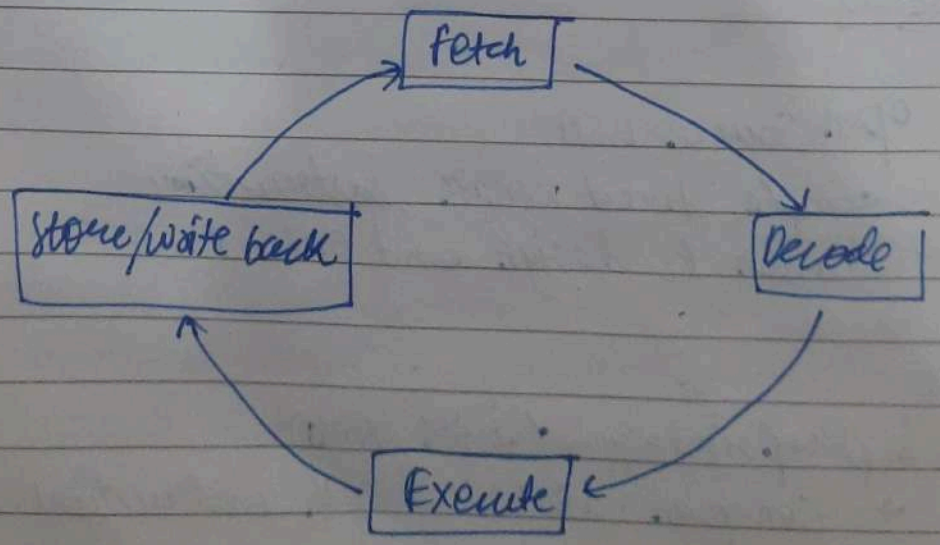


Explanation

→ PC holds next instruction address; IR stores current instruction

→ MAR & MDR handle memory access, ACC stores intermediate results.

Ans 2



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- CPU fetches instructions, decode it, execute it and write back.

Ans 5:-

opcode	operands

Explanation

- opcode specifies the operations
- operands specify data or memory location

Ans 4

Feature	RISC	CISC
instruction	simple	complex
length	fixed	variable
speed	fast	slower
memory	more usage	less usage.

Explanation

→ RISC focuses on simplicity and speed
- CISC focuses on powerful and complexity.

Section - C

Ans 1- Impact on performance are:-

- Memory access is slower than register access
- increases CPI due to memory latency.
- overall execution time increases → performance decreases.

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Ans2 The should choose RISC architecture because

- simple and fixed length instructions
- Easier to pipeline efficiency.
- Require simpler hardware design.

Ans3 CPU executes this instruction as:-

Step1:- Fetch

Instruction is fetched from memory into IR.

Step2:- Decode

- opcode is decoded and registers are identified.

Step3:- Execute

- Data is read from registers.
→ ALU perform operation
→ Result is stored back in register.

Ans4 x86 is widely used despite complexity because.

→ Backward compatibility with older software.
→ Large ecosystem (OS, applications, industry, support)
→ Modern CPU internally optimized.
→ Despite complexity compatibity and ecosystem keep x86 dominant.

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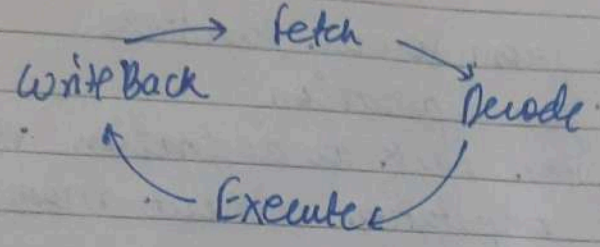
Section-D

Ans1)



Explanation of fetch stage

- PC holds address of next instructions
- address sent to memory via MAR.
- instruction fetched into MDR and loaded into IR
- PC is incremented to next instruction

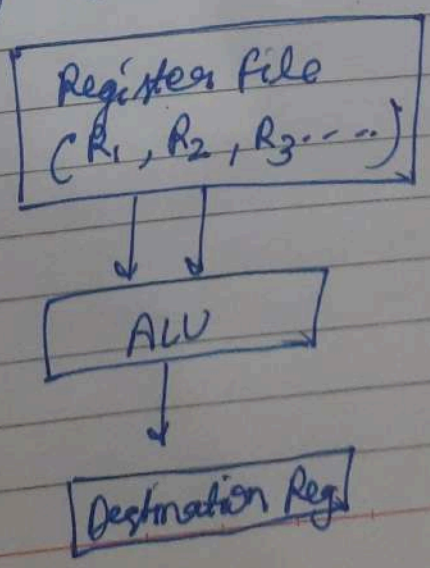
Ans2

funct 7	rs 2	rs 1	function3	rd	opcode

Explanation

→ opcode - specifies operation type
rd - Destination register
rs1, rs2 - Source Register
function3 & funct 7 - Define exact operation
fixed - length (32 bit) format simplies decoding.

Ans3



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Explanation

- Data is read from source registers
- Sent to ALU for operation
- Result is written back to destination register.
- Enable fast computation without memory access.